8. Write a C program for monoalphabetic cipher is that both sender and receiver must commit the permuted cipher sequence to memory. A common technique for avoiding this is to use a keyword from which the cipher sequence can be generated. For example, using the keyword CIPHER, write out the keyword followed by unused letters in normal order and match this against the plaintext letters:

plain: a b c d e f g h i j k l m n o p q r s t u v w x y z

cipher: C I P H E R A B D F G J K L M N O Q S T U V W X Y Z

Aim:

To implement a monoalphabetic substitution cipher using the keyword CIPHER**.**

Procedure:

1. Start the program.
2. Read the plaintext.
3. Construct the substitution alphabet using the keyword CIPHER, followed by unused letters.
4. Replace each plaintext letter with its corresponding ciphertext letter.
5. Display the encrypted text**.**

Code:

```
#include <stdio.h>
#include <ctype.h>
int main() {
   char text[100];
   char key[] = "CIPHERABDFGJKLMNOQSTUVWXYZ";
   int i, x;
   printf("Enter plaintext: ");
   fgets(text, sizeof(text), stdin);
```

```
    for (i = 0; text[i]; i++) {
        if (isalpha((unsigned char)text[i])) {
            x = toupper(text[i]) - 'A';
            text[i] = key[x];
        }
    }
    printf("Ciphertext: %s", text);
    return 0;
}
```

Input:

Enter plaintext:

HELLO

Output:

